Proposition Examples and Functions

1.65 To describe the various restaurants in the city, we let p denote the statement, "The food is good," q denote the statement, "The service is good," and r denote the statement, "The rating is three-star." Write the following statements in symbolic form.

(a) Either the food is good, or the service is good, or both. – $(p \vee q)$
(b) Either the food is good or the service is good, but not both. –
(c) The food is good while the service is poor.
(d) It is not the case that [both the food is good and the rating is three-star.]
(e) If both the food and services are good, then the rating will be three-star.
(f) It is not true that a [three-star rating always means good food and good service.]

1. $p \vee q$
2. $p \oplus q$
3. $p \wedge \overline{q}$
4. $\overline{(p \wedge r)}$
5. $(p \wedge q) \rightarrow r$
6. $\overline{(r \rightarrow p \wedge q)}$

Example 1.16 An island has two tribes of natives. Any native from the first tribe always tells the truth, while any native from the other tribe always lies. You arrive at the island and ask a native if there is gold on the island. He answers, "There is gold on the island if and only if I always tell the truth." Which tribe is he from? Is there gold on the island? As it turns out, we

1. Can not be Concluded. 2. Yes.

p = The native is always Speaking Truth.
q = There is gold on island

The answer of tribe = $p \leftrightarrow q$

	p	q	$p \leftrightarrow q$
Tribe-1 –	T	T	T
Tribe-2 –	F	T	F

~ From above it Can be Concluded that q is true Under all Circumstances.
⇒ There is gold Present on the island.

~ Syllabus : Topics up to 1.10.

Ch. 3

Relations and Functions.

~ $S = \{ \underset{1}{\underline{abc}}, \underset{2}{\underline{acd}}, xyz, \underset{3}{\underline{axz}} \}$

- Element 1, 2, 3 are related withe each other their names have Common first letter (a)

~ $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$

Relation: 2, 4, 6, 8 are related. (even No.)

1, 2 are not related.

* Consider two Sets

$A = \{a, b\}$, $B = \{a, c, d\}$

- Cartesian Product

$A \times B = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d)\}$

- Binary relation from A to B is Subset of $A \times B$

$A = \{1, 2\}$, $B = \{1, 3\}$

$A \times B = \{(1,1), (1,3), (2,1), (2,3)\}$

"Less than Relation"

$R = \{(1,3), (2,3)\}$ $\quad\quad R \subset (A \times B)$

$(1,1) \in R$?

is $1 < 1$? - No. $(1,1) \notin R$

~ Relation is Set of ordered Pairs.

~ (a,b) and (b,a) are different.

~ (a,a) is also Valid Pair.

* Graphical Form of Relationship

$A = \{a, b, c, d\}$ $\quad B = \{\alpha, \beta, \gamma\}$

$R = \{(a,\alpha), (b,\gamma), (c,\alpha), (c,\gamma), (d,\beta)\}$

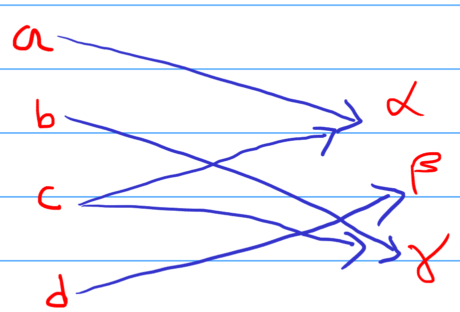
where R is Binary Relation from A to B

~ Tabular form

	α	β	γ
a	✓		
b			✓
c	✓		✓
d		✓	

(Relation Matrix)

~ Graph:



Directed Graph

~ Relation operations

1. $R1 \cup R2$
2. $R1 \cap R2$
3. $R1 - R2$
4. $R1 \oplus R2$

~ $R1$ is on $A \times B$ where
A = Student Names
B = Subject List
$(a,b) \in R1 \Rightarrow$ The subject which student has taken.

~ $R2$ is on $A \times B$ where (as above)
$(a,b) \in R2 \Rightarrow$ The student a likes subject B.

* Ternary Relationship among A, B and C is defined as subset of Cartesian Product of $(A \times B)$ and C
$(A \times B) \times C$

$A = \{a, b\}$ $B = \{\alpha, \beta\}$, $C = \{1, 2\}$

$A \times B = \{(a,\alpha), (a,\beta), (b,\alpha), (b,\beta)\} \times \{1, 2\}$

$(A \times B) \times C = \{((a,\alpha),1), ((a,\alpha),2), ((a,\beta),1), ((a,\beta),2), ((b,\alpha),1), ((b,\alpha),2), ((b,\beta),1), ((b,\beta),2)\}$

~ Ternary Relation R on A, B & C is
$R \subset (A \times B) \times C$